

EXP.NO: 2 Write JavaScript Programme to validate the following field the
HTMLRegistration page.

AIM:

To write JavaScript code to validate user input in a registration form. The validation will ensure that the data entered by the user meets specific criteria for each field. This will help in maintaining data integrity and prevent invalid submissions.

ALGORITHM:

1. Input:

- The user fills in the following fields: First Name, Password, E-mail, Mobile Number, Last Name, and Address.

2. Validation Steps:

1. Validate First Name:

- o Check if the first name contains only alphabets.
- o Ensure the length of the First Name is at least 6 characters.
- o If any of these conditions fail, display an error message indicating the required input.

2. Validate Password:

- o Check if the password is at least 6 characters long.
- o If it is less than 6 characters, display an error message.

3. Validate E-mail:

- o Use a regular expression to check if the email follows the standard pattern:
name@domain.com.
- o Ensure that the email contains no invalid characters and matches the required pattern.

- o If it doesn't match, display an error message indicating an invalid email

format.

4. Validate Mobile Number:

- o Check if the mobile number contains exactly 10 digits.
- o Ensure there are no alphabets or special characters.
- o If the number is invalid, display an error message indicating the correct format.

5. Validate Last Name:

- o Ensure the Last Name field is not empty.
- o If it is empty, display an error message.

6. Validate Address:

- o Ensure the Address field is not empty.
- o If it is empty, display an error message.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Registry System</title>

  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0/css/all.min.css">

  <style>

    * { margin: 0; padding: 0; box-sizing: border-box; font-family: 'Helvetica', 'Arial', sans-serif; }

    body {

      background-color: #ffffff;
```

```
color: #000000;  
padding: 50px 20px;  
display: flex;  
flex-direction: column;  
align-items: center;  
min-height: 100vh;  
}
```

```
.container {  
    border: 3px solid #000000;  
    padding: 40px;  
    width: 100%;  
    max-width: 500px;  
    background: #ffffff;  
    margin-bottom: 60px;  
    box-shadow: 15px 15px 0px #000000;  
}
```

```
h2 {  
    text-align: center;  
    margin-bottom: 40px;  
    font-weight: 900;  
    letter-spacing: 5px;  
    text-transform: uppercase;  
    border-bottom: 3px solid #000;
```

```
padding-bottom: 10px;  
}
```

```
.form-group { margin-bottom: 25px; position: relative; }
```

```
label {  
    display: block;  
    margin-bottom: 8px;  
    font-size: 0.75em;  
    font-weight: 800;  
    text-transform: uppercase;  
    letter-spacing: 1px;  
}
```

```
input, textarea {  
    width: 100%;  
    padding: 12px;  
    border: 2px solid #000;  
    background: #fff;  
    color: #000;  
    outline: none;  
    font-weight: 600;  
}
```

```
input:focus, textarea:focus { background: #f0f0f0; }
```

```
.error-text { color: #ff0000; font-size: 0.7em; margin-top: 5px; display: none; font-weight: bold; }
```

```
.invalid { border: 2px solid #ff0000 !important; }
```

```
.pwd-toggle { position: absolute; right: 15px; top: 38px; cursor: pointer; color: #000; }
```

```
.btn-main {  
    width: 100%;  
    padding: 18px;  
    border: 3px solid #000;  
    background: #000;  
    color: #fff;  
    font-weight: 900;  
    cursor: pointer;  
    text-transform: uppercase;  
    letter-spacing: 2px;  
    transition: 0.2s;  
}
```

```
.btn-main:hover { background: #fff; color: #000; }
```

```
.table-section { width: 100%; max-width: 900px; }
```

```
.table-header {
```

```
display: flex;

justify-content: space-between;

align-items: center;

border-bottom: 3px solid #000;

padding-bottom: 15px;

margin-bottom: 20px;
}
```

```
.action-buttons { display: flex; gap: 10px; }
```

```
.btn-util {

background: #fff;

color: #000;

border: 2px solid #000;

padding: 8px 15px;

cursor: pointer;

font-size: 0.7em;

font-weight: 800;

text-transform: uppercase;
}
```

```
.btn-util:hover { background: #000; color: #fff; }
```

```
table { width: 100%; border-collapse: collapse; }
```

```
th, td { border: 2px solid #000; padding: 12px; text-align: left; font-size: 0.85em; }
```

```
th { background-color: #000; color: #fff; text-transform: uppercase; }
```

```
tr:nth-child(even) { background-color: #f9f9f9; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Registration</h2>
```

```
<form id="regForm" novalidate>
```

```
<div class="form-group">
```

```
<label>First Name</label>
```

```
<input type="text" id="firstName" placeholder="Min 6 alphabets">
```

```
<div id="firstNameError" class="error-text">Required: Min 6 Alphabets</div>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Last Name</label>
```

```
<input type="text" id="lastName" placeholder="Required">
```

```
<div id="lastNameError" class="error-text">Required Field</div>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>Password</label>
```

```
<input type="password" id="password" placeholder="Min 6 characters">
```

```
<i class="fa-solid fa-eye pwd-toggle" id="togglePass"></i>

<div id="passwordError" class="error-text">Must be 6+ characters</div>

</div>

<div class="form-group">

  <label>Email</label>

  <input type="email" id="email" placeholder="name@domain.com">

  <div id="emailError" class="error-text">Invalid Email Address</div>

</div>

<div class="form-group">

  <label>Mobile Number</label>

  <input type="text" id="mobile" maxlength="10" placeholder="10 Digits">

  <div id="mobileError" class="error-text">10 Digits Required</div>

</div>

<div class="form-group">

  <label>Address</label>

  <textarea id="address" rows="2" placeholder="Full Details"></textarea>

  <div id="addressError" class="error-text">Required Field</div>

</div>

<button type="submit" class="btn-main">Submit & Save</button>

</form>

</div>

<div class="table-section">

  <div class="table-header">

    <h3>Stored Records</h3>
```



```
<div class="action-buttons">

    <button class="btn-util" onclick="exportCSV()">Export CSV</button>

    <button class="btn-util" style="border-color: red; color: red;" onclick="clearData()">Clear
All</button>

</div>

</div>

<table id="userTable">

    <thead>

        <tr>

            <th>Name</th>

            <th>Email</th>

            <th>Mobile</th>

            <th>Address</th>

        </tr>

    </thead>

    <tbody id="tableBody"></tbody>

</table>

</div>
```

```
<script>

const form = document.getElementById('regForm');

const tableBody = document.getElementById('tableBody');

const patterns = {

    firstName: /^[A-Za-z]{6,}$/,

    email: /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/,

    mobile: /^[0-9]{10}$/
```

```
};
```

```
function loadTable() {  
  
    const data = JSON.parse(localStorage.getItem('user_registry')) || [];  
  
    tableBody.innerHTML = "";  
  
    data.forEach(user => {  
  
        const row = `<tr>  
  
            <td>${user.firstName} ${user.lastName}</td>  
  
            <td>${user.email}</td>  
  
            <td>${user.mobile}</td>  
  
            <td>${user.address}</td>  
  
        </tr>`;   
  
        tableBody.innerHTML += row;  
  
    });  
  
}
```

```
function exportCSV() {  
  
    const data = JSON.parse(localStorage.getItem('user_registry')) || [];  
  
    if (data.length === 0) return alert("No data to export");  
  
    let csvContent = "data:text/csv;charset=utf-8,Name,Email,Moblie,Address\n";  
  
    data.forEach(u => {  
  
        csvContent += `${u.firstName} ${u.lastName},${u.email},${u.mobile},"${u.address.replace(/\n/g, "  
")}"\n`;  
  
    });  
  
}
```

```
const encodedUri = encodeURIComponent(csvContent);

const link = document.createElement("a");

link.setAttribute("href", encodedUri);

link.setAttribute("download", "registry_data.csv");

document.body.appendChild(link);

link.click();

}
```

```
function clearData() {

    if(confirm("Permanently delete all records?")) {

        localStorage.removeItem('user_registry');

        loadTable();

    }

}
```

```
function validate(input, regex) {

    const errorDiv = document.getElementById(input.id + 'Error');

    let isValid = (input.id === 'lastName' || input.id === 'address')

        ? input.value.trim().length > 0

        : (input.id === 'password' ? input.value.length >= 6 : regex.test(input.value));

    input.classList.toggle('invalid', !isValid);

    errorDiv.style.display = isValid ? 'none' : 'block';

    return isValid;

}
```

```
document.getElementById('togglePass').addEventListener('click', function() {  
    form.password.type = (form.password.type === 'password') ? 'text' : 'password';  
    this.classList.toggle('fa-eye-slash');  
});
```

```
form.addEventListener('submit', (e) => {  
    e.preventDefault();  
  
    const v1 = validate(form.firstName, patterns.firstName);  
    const v2 = validate(form.lastName, null);  
    const v3 = validate(form.password, null);  
    const v4 = validate(form.email, patterns.email);  
    const v5 = validate(form.mobile, patterns.mobile);  
    const v6 = validate(form.address, null);
```

```
if (v1 && v2 && v3 && v4 && v5 && v6) {
```

```
    const userData = {  
        firstName: form.firstName.value,  
        lastName: form.lastName.value,  
        email: form.email.value,  
        mobile: form.mobile.value,  
        address: form.address.value  
    };
```

```
    let storedData = JSON.parse(localStorage.getItem('user_registry')) || [];  
    storedData.push(userData);
```

```

        localStorage.setItem('user_registry', JSON.stringify(storedData));

        alert("SUCCESS: DATA COMMITTED.");

        form.reset();

        window.location.reload();

    }

});

window.onload = loadTable;

</script>
</body>
</html>

```

Output:

REGISTRATION

FIRST NAME

LAST NAME

PASSWORD

EMAIL

MOBILE NUMBER

ADDRESS

SUBMIT & SAVE

Stored Records

EXPORT CSV

CLEAR ALL

NAME	EMAIL	MOBILE	ADDRESS
Prasanth HarishKumar	241501147@rajalakshmi.edu.in	8122950405	kattupakkam

RESULT:

HTML page with Form is created and is validated using Javascript